Data Directory

This directory contains the datasets used for training and evaluating PromptGFM-Bio.

■■ **Important**: Large data files are excluded from version control. You must download them separately.

Directory Structure

```
data/ ■■■ raw/ # Raw downloaded datasets (~10GB) ■ ■■■ string/ # STRING
protein-protein interactions ■ ■■■ biogrid/ # BioGRID interaction data ■ ■■■
disgenet/ # DisGeNET gene-disease associations ■ ■■■ hpo/ # Human Phenotype
Ontology ■ ■■■ orphanet/ # Orphanet rare disease data ■■■ processed/ #
Preprocessed data (~2GB) ■ ■■■ biomedical_graph.pt # Main knowledge graph ■ ■■■
gene_disease_edges.csv # Processed associations ■ ■■■ graph_stats.txt #
Statistics ■■■ splits/ # Train/val/test splits ■■■ train.csv ■■■ val.csv ■■■
test.csv
```

Downloading Data

Automatic Download (Recommended)

Use the provided download script to fetch all required datasets:

```
# Download all datasets python scripts/download_data.py --all # Or download
specific datasets python scripts/download_data.py --datasets string biogrid
disgenet hpo orphanet
```

Manual Download

If automatic download fails, you can manually download from these sources:

1. STRING Database

- **URL**: https://string-db.org/cgi/download
- **File**: `9606.protein.links.v12.0.txt.gz` (Human protein links)
- **Size**: ~4GB
- **Place in**: `data/raw/string/`

2. BioGRID

- **URL**: https://downloads.thebiogrid.org/BioGRID/Release-Archive/
- **File**: `BIOGRID-ALL-4.4.224.tab3.txt` (or latest version)
- **Size**: ~500MB
- **Place in**: `data/raw/biogrid/`

3. DisGeNET

- **URL**: https://www.disgenet.org/downloads
- **File**: `all_gene_disease_associations.tsv.gz`
- **Size**: ~200MB
- **Note**: Requires free registration
- **Place in**: `data/raw/disgenet/`

4. Human Phenotype Ontology (HPO)

- **URL**: https://hpo.jax.org/app/download/annotation
- **Files**:
- `genes_to_phenotype.txt`
- `phenotype_to_genes.txt`
- `phenotype.hpoa`
- **Size**: ~50MB
- **Place in**: `data/raw/hpo/`

5. Orphanet

- **URL**: http://www.orphadata.org/cgi-bin/index.php
- **Files**:
- `en_product1.xml` (Rare diseases and classifications)
- `en_product4.xml` (Genes associated with rare diseases)
- `en_product6.xml` (Epidemiological data)
- **Size**: ~100MB
- **Place in**: `data/raw/orphanet/`

Preprocessing Data

After downloading raw data, preprocess it to build the knowledge graph:

```
# Run preprocessing pipeline python scripts/preprocess_all.py # This will: # 1. Parse raw data files # 2. Map gene/protein identifiers # 3. Construct heterogeneous graph # 4. Generate train/val/test splits # 5. Save processed data to data/processed/
```

Expected output:
- `data/processed/biomedical_graph.pt` - PyTorch Geometric graph object
- `data/processed/merged_gene_disease_edges.csv` - Processed associations
- `data/processed/biomedical_graph_stats.txt` - Graph statistics

Data Statistics

After preprocessing, you should have:

Component	Count
Genes/Proteins	~20,000 nodes
Diseases	~10,000 nodes
Phenotypes	~15,000 nodes
Protein-Protein Interactions	~9.7M edges
Gene-Disease Associations	~87K edges
Gene-Phenotype Mappings	~200K edges

Disk Space Requirements

- **Raw data**: ~10 GB
- **Processed data**: ~2 GB

- **Temporary files (during preprocessing)**: ~3 GB
- **Total**: ~15 GB

Ensure you have sufficient disk space before downloading.

Data Versions

The current implementation uses:

- **STRING v12.0** (2021)
- **BioGRID 4.4.224** (2023)
- **DisGeNET v7.0** (2020)
- **HPO** (monthly releases)
- **Orphanet** (updated quarterly)

To update to newer versions:
1. Download new files to `data/raw/`
2. Update file paths in `src/data/download.py`
3. Re-run preprocessing

Data License and Attribution

All datasets are publicly available but have specific licenses:

- **STRING**: Creative Commons BY 4.0
- **BioGRID**: MIT License
- **DisGeNET**: ODbL 1.0 (Open Database License)
- **HPO**: Free for research use
- **Orphanet**: Creative Commons BY 4.0

Important: If you use this data in publications, cite the original sources (see LICENSE).

Troubleshooting

Download Failures

If downloads fail:
1. Check internet connection
2. Verify URLs are still valid
3. Try manual download (links above)
4. Check if registration is required (DisGeNET)

Preprocessing Errors

Common issues:
- **Missing files**: Ensure all raw data is downloaded
- **Parsing errors**: Check file formats match expected versions
- **Memory errors**: Preprocessing requires ~16GB RAM
- **Disk space**: Ensure sufficient space for temporary files

See docs/TROUBLESHOOTING.md for more help.

Using Your Own Data

To integrate additional datasets:

1. **Add downloader**: Modify `src/data/download.py`
2. **Add parser**: Extend `src/data/preprocess.py`
3. **Update graph construction**: Modify `build_biomedical_graph()`
4. **Test**: Verify graph structure and statistics

Example:

```
# In src/data/preprocess.py def parse_custom_dataset(file_path): df =
pd.read_csv(file_path) # Process and return edges return edges
```

Data Privacy and Ethics

This project uses only publicly available biomedical data:
- No patient-level information
- No protected health information (PHI)
- De-identified, aggregated data only

For clinical applications: Additional validation and ethical review required. This software is for research purposes only.

For more information on data handling, see:
- ARCHITECTURE.md - Data pipeline details
- SETUP.md - Complete setup instructions
- scripts/download_data.py - Download script source